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LAB 07 INSTRUCTIONS

For

Build a filter evidence matrix

Course Code: C1123

Conducted by

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Version 5.1

Lab 07 — Build a filter evidence matrix

Course: Wireshark Network Analysis Masterclass (C1123)

Topic: 07 — Display Filters

Learning outcome: LO2 — Navigate, filter, summarize and time network exchanges.

Time: about 30 minutes

Version: v5.1 · 5 October 2026

Scenario

The team keeps sharing filters that silently hide evidence. You build a tested filter matrix so everyone answers the same question the same way.

Goal

Parentheses make mixed and/or expressions explicit.

What you will produce

A filter evidence matrix with verified frame numbers.

Before you start

- Wireshark 4.6 or later, with the TShark command-line tools on PATH.
- Python 3 for the verification and export scripts (Windows: use `py -3` in place of `python3`).
- Open this lab folder in a terminal; every file the lab needs is inside it.
- Use only the supplied synthetic captures — do not capture on a network you are not authorised to monitor.

Files for this lab

File	What it is for
data/branch-office.pcap	Synthetic branch-office capture used by the lab
assets/scenario.md	The help-desk ticket that sets the scenario
assets/checks.json	Filters and frame counts the fixture must satisfy
assets/expected-evidence.png	The packet list your filter should produce
scripts/verify.py	Checks the capture facts with TShark
scripts/export_evidence.py	Exports this lab's evidence rows to outputs/evidence.csv
outputs/findings.md	Findings sheet you complete
assets/filter-matrix.csv	Filter matrix template (question, filter, frames)

Lab at a glance

1. Find the NXDOMAIN response and SYNs
2. Compare two bracketings of one filter
3. Test `!(ip.addr == x)` vs `ip.addr != x`
4. Save five verified filters with frames

Step-by-step

1. Prepare your lab workspace. Open this lab folder in a terminal. Windows uses `py -3` in place of `python3`. Wireshark GUI alone is sufficient for the investigation; CLI scripts require TShark on PATH. Run the fixture verification first.

```
python3 scripts/verify.py
```

2. Open `data/branch-office.pcap`. Apply `dns.flags.rcode == 3` and record the returned response frame. Clear the filter and apply `tcp.flags.syn == 1`.

3. Compare `(ip.src == 192.0.2.10 && udp.port == 53) || tcp.port == 80` with `ip.src == 192.0.2.10 && (udp.port == 53 || tcp.port == 80)`. Record both counts and the direction excluded by the second expression.

4. Apply `!(ip.addr == 192.0.2.53)`. Verify that DNS packets disappear; compare `ip.addr != 192.0.2.53` on Wireshark 4.6 or later. Record the actual version and equality behaviour.

5. Save five useful filters in `outputs/filters.txt`. For each, state the question, filter and matching frame numbers; compile them with `scripts/verify.py`.

6. Export a reproducible packet table. Record the filter, frame numbers, measurement, explanation and limitation in `outputs/findings.md`.

```
python3 scripts/export_evidence.py
```

Expected evidence

With the lab filter applied, your packet list should match the frames below (produced by TShark from this lab's own capture).

dns.flags.rcode == 3					
No.	Time	Source	Destination	Protocol	Info
6	0.070000	192.0.2.53	192.0.2.10	DNS	Standard query response 0x0066 No such name A mis...
branch-office.pcap Displayed: 1					

Lab 07 expected evidence

Test it

The supplied `dns.flags.rcode == 3` expression matches 1 frame(s) in the specified capture. Run `scripts/verify.py` and compare the listed frame numbers. Keep your findings and exported table in `outputs/`. Explain the observed result rather than only copying a count.

Troubleshooting

- TShark not found: install Wireshark CLI tools and add the installation folder to PATH; on Windows use the Wireshark install directory.
- A filter returns zero: clear other filters, use the specified capture, and check the expression is in the display toolbar.
- TLS remains opaque: select the matching `lab-tls.keys` file by absolute path, reload, and remove a key-log preference from another lab.

Try it with TShark

The same evidence from the command line — run it from this lab folder:

```
tshark -n -r data/branch-office.pcap -Y "dns.flags.rcode == 3"
```

Challenge

Develop and explain an alternative filter.

Reflection

Which second observation point would strengthen your conclusion?

Extension (optional)

Rewrite two of your matrix filters with the set (in {...}) and matches operators and confirm the frame lists are unchanged.

Reset

Clear all display filters and return to the C1123-Analyst or Default profile. If you loaded a TLS key log, remove it from Preferences > Protocols > TLS. The supplied captures never need regenerating for the core lab.

Note: The same steps appear in the Learner Guide. The slides show only the scenario and a summary.